
Technical Report

for

ESP32-Based Smart Cane for the Blind and Visually Impaired

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Abstract

Visual impairment significantly affects an individual's ability to navigate safely and independently, with traditional white canes providing limited detection of obstacles beyond ground level and offering no emergency communication capabilities. This project presents the design and implementation of an ESP32-Based Smart Cane for the Blind and Visually Impaired, an assistive mobility device that enhances the functionality of the conventional white cane through embedded sensing, intelligent processing, and real-time feedback. The prototype integrates dual ultrasonic sensors to detect both ground-level and upper-body obstacles, a vibration motor to provide haptic feedback proportional to obstacle proximity, a DFPlayer Mini module for voice guidance, and an MPU6050 inertial measurement unit for fall detection. In the event of a confirmed fall, the system utilizes the ESP32's Wi-Fi capability to send an emergency notification through an IFTTT webhook, enabling caregivers to respond promptly. The system is designed to provide continuous environmental awareness while remaining affordable and portable for everyday use. The developed prototype successfully demonstrates reliable obstacle detection and classification, adaptive haptic feedback, voice-based distance warnings, and automatic fall detection with remote notification, providing a practical proof of concept for an intelligent mobility aid. Although the current implementation has limitations, including the absence of downward-facing obstacle detection, battery monitoring, and GPS-based location tracking, it establishes a strong foundation for future enhancements. By combining low-cost hardware with embedded intelligence, the smart cane has the potential to improve the safety, independence, and confidence of blind and visually impaired individuals during navigation. The prototype integrates dual ultrasonic obstacle detection, haptic feedback, voice guidance, and Wi-Fi-based fall notification as its core functionality.

1 Introduction

1.1 *This technical report presents the design, development, implementation, and evaluation of the ESP32-Based Smart Cane for the Blind and Visually Impaired. The report documents the engineering process followed by the project team, including the problem being addressed, the objectives of the project, the system requirements, hardware and software design, implementation approach, and the functionality of the completed prototype. It also highlights the technologies and components used to develop the smart cane and explains how they work together to provide an intelligent mobility aid for visually impaired users. Furthermore, this report serves as a record of the project's outcomes and provides an assessment of the prototype's capabilities, limitations, and potential areas for future improvement. It is intended to demonstrate the practical application of embedded systems, sensors, and Internet of Things (IoT) technologies in solving real-world accessibility challenges while documenting the knowledge, skills, and engineering practices applied throughout the development process as part of the CSC 1304 Practical Skills Development course.*

1.2 User Challenge

People who are blind or visually impaired face significant challenges when navigating unfamiliar environments independently. While the traditional white cane is an essential mobility aid, it primarily detects obstacles at ground level and within the cane's physical reach. As a result, users remain vulnerable to obstacles positioned above waist level, such as signboards, tree branches, poles, and parked vehicles, which may not be detected before causing injury. Additionally, conventional canes do not provide any mechanism for detecting falls or notifying caregivers during emergencies, leaving users at greater risk when traveling alone. The ESP32-Based Smart Cane for the Blind and Visually Impaired was developed to address these limitations by incorporating intelligent sensing and communication technologies into a traditional mobility aid. The system uses dual ultrasonic sensors to detect obstacles at different heights, a vibration motor and voice prompts to provide immediate feedback, and an MPU6050 sensor to detect falls. In the event of a confirmed fall, the cane automatically sends an emergency notification to a designated caregiver through a Wi-Fi connection. By enhancing environmental awareness and enabling emergency communication, the smart cane aims to improve the safety, independence, and confidence of visually impaired individuals during daily navigation.

1.3 Project Goals

Build an affordable, reliable smart mobility aid.

Detect obstacles at both ground and upper-body height.

Deliver real-time haptic and voice feedback.

Automatically detect falls and notify a caregiver over Wi-Fi.

1.4 *The ESP32-Based Smart Cane for the Blind and Visually Impaired is an intelligent assistive mobility device designed to improve the safety and independence of individuals with visual impairments. The project enhances the functionality of the traditional white cane by integrating embedded sensors, real-time processing, and wireless communication technologies. The system employs dual ultrasonic sensors to detect obstacles at different heights, a vibration motor and voice feedback module to provide immediate navigation assistance, and an MPU6050 sensor to detect falls. When a fall is confirmed, the ESP32 microcontroller sends an emergency notification to a caregiver through a Wi-Fi connection using an IFTTT webhook. The primary purpose of the project is to provide users with greater environmental awareness while enabling rapid assistance during emergency situations. The project's main objectives are to develop an affordable and reliable smart mobility aid, improve obstacle detection beyond the capabilities of a conventional white cane, and enhance user safety through automatic fall detection and emergency notification. Its innovative design combines multiple assistive technologies into a single low-cost device that provides both haptic and voice-based feedback, allowing users to receive timely information without relying on visual cues. By detecting obstacles at both ground and upper-body levels and providing remote emergency alerts, the smart cane directly addresses the mobility and safety challenges faced by visually impaired individuals, enabling them to navigate their surroundings with increased confidence and independence.*

1.5 Functional Requirements

The Smart Cane is designed to provide intelligent navigation assistance and emergency support for visually impaired users. Its functional requirements are grouped into several functional areas based on the services performed by the system.

1.5.1 Obstacle Detection. The primary function of the smart cane is to detect obstacles that may hinder the user's movement. The system uses two ultrasonic sensors mounted at different heights to detect both ground-level and upper-body obstacles. The obstacle detection functions include: Detect obstacles within a predefined distance from the user. Differentiate between low obstacles and taller obstacles using the two ultrasonic sensors. Continuously measure the distance between the cane and nearby objects in real time. Update obstacle information as the user moves to ensure timely warnings.

1.5.2 User Feedback and Navigation Assistance. The system provides immediate feedback to the user whenever an obstacle is detected. Feedback is delivered through vibration and voice prompts to enable safe navigation without requiring visual interaction. The user feedback functions include: Activate the vibration motor when an obstacle is detected. Increase vibration intensity as the user approaches an obstacle. Play voice warnings indicating the presence and approximate distance of obstacles. Prevent repeated or overlapping voice messages while an audio alert is already playing. Provide continuous feedback until the obstacle is no longer within the warning range.

1.5.3 Fall Detection. To improve user safety, the smart cane continuously monitors its orientation and movement using an MPU6050 accelerometer and gyroscope. The system identifies abnormal movements that may indicate a fall. The fall detection functions include: Continuously monitor acceleration and orientation data. Detect sudden impacts followed by abnormal tilt angles. Confirm whether a fall has occurred using predefined thresholds. Reset the fall detection state automatically after the user resumes normal movement.

1.5.4 Emergency Notification Following confirmation of a fall, the smart cane automatically sends an emergency alert to notify a caregiver or emergency contact. The emergency notification functions include: Connect to a configured Wi-Fi network. Trigger an IFTTT webhook after a confirmed fall. Send an emergency notification requesting immediate assistance. Retry transmission if the first notification attempt fails. Prevent multiple notifications for the same fall event.

1.5.5 System Control and Processing. The ESP32 microcontroller coordinates all sensing, processing, and communication activities within the smart cane. The system control functions include: Initialize all hardware components during startup. Read data continuously from the ultrasonic sensors and MPU6050. Process sensor readings in real time. Coordinate obstacle detection, audio output, vibration feedback, and fall detection simultaneously. Manage power-up initialization and communication between all connected modules.

1.5.6 Audio Management. The smart cane uses a DFPlayer Mini module to provide clear spoken guidance to the user. The audio management functions include: Play pre-recorded voice messages stored on a microSD card. Select the appropriate voice prompt based on obstacle distance. Ensure that only one audio message is played at a time. Resume obstacle monitoring immediately after audio playback finishes.

1.5.7 Reliability and Safety. The system is designed to operate reliably during normal usage while ensuring consistent assistance for the user. The reliability functions include: Continuously monitor sensor data throughout operation. Ignore invalid or out-of-range sensor readings where possible. Maintain obstacle detection while other system functions execute. Automatically recover normal operation after a fall event has been resolved. These functional requirements collectively enable the Smart Cane to provide intelligent obstacle detection, real-time user feedback, automatic fall detection, and emergency communication, thereby improving the safety, independence, and confidence of blind and visually impaired users during navigation.

2 Project Results

2.1 Product Design

The ESP32-Based Smart Cane is designed as an embedded Internet of Things (IoT) assistive device that integrates sensing, processing, user feedback, and wireless communication into a single portable mobility aid for blind and visually impaired users.

The ESP32 microcontroller serves as the central processing unit, receiving data from multiple sensors, processing the information in real time, and controlling the appropriate output devices. The input subsystem consists of two ultrasonic sensors positioned at different heights to detect obstacles at both ground level and upper-body level. An MPU6050 accelerometer and gyroscope continuously monitors the orientation and movement of the cane to detect falls or abnormal motion. A push button is also included to power the system and provide user control. Power is supplied by a 3.7 V lithium-ion rechargeable battery, which is connected to a voltage regulation module that provides stable operating voltage to the ESP32 and all peripheral components.

The ESP32 microcontroller acts as the brain of the system. It continuously acquires sensor readings, processes obstacle distances, determines whether a fall has occurred, and decides which feedback mechanism should be activated. It also manages Wi-Fi communication for emergency notifications. The feedback subsystem provides multiple forms of user assistance. A vibration motor generates haptic feedback whose intensity increases as obstacles become closer, allowing users to estimate obstacle proximity through touch. A DFPlayer Mini voice module connected to a speaker plays prerecorded voice messages that announce obstacle distance and navigation warnings. In addition, a buzzer produces short audible alerts to immediately notify the user whenever obstacles are detected or other warning conditions occur.

For emergency situations, the ESP32 utilizes its built-in Wi-Fi capability to communicate with an IFTTT webhook. When the MPU6050 confirms that the user has fallen, the ESP32 automatically sends an emergency notification through the internet to a registered caregiver or guardian, enabling rapid assistance.

The overall system follows a straightforward data flow. Sensor data is collected by the ESP32, processed using embedded software, and translated into appropriate outputs such as vibration, buzzer alerts, voice guidance through the speaker, or emergency notifications. By combining obstacle detection, fall detection, audio guidance, haptic feedback, and wireless communication into a single integrated architecture, the smart cane enhances the safety, mobility, and independence of visually impaired users while remaining affordable and practical for everyday use.

2.2 Product Functionality and Screenshots

The ESP32-Based Smart Cane for the Blind and Visually Impaired was designed as an embedded assistive system that combines sensing, processing, user feedback, and wireless communication into a single portable device. The ESP32 microcontroller acts as the central processing unit, collecting information from multiple sensors, processing the data in real time, and controlling the output devices. The system provides obstacle detection, navigation assistance, fall detection, and emergency notification to improve the mobility and safety of visually impaired users. The hardware architecture consists of two ultrasonic sensors, an MPU6050 accelerometer and gyroscope module, a DFPlayer Mini audio module, a vibration motor, and an ESP32 microcontroller. The ESP32 continuously reads data from the sensors, determines whether an obstacle or fall has occurred, and activates the appropriate feedback mechanism. Audio warnings and vibration alerts provide immediate guidance to the user, while the built-in Wi-Fi capability enables emergency notifications to caregivers through an IFTTT webhook after a confirmed fall.

2.3 Project website and repository

Repo: <https://github.com/maurice64bit/smartcane.git>

Website: <https://maurice64bit.github.io/smartcane>

3 Limitations and Next Steps

3.1 Limitations

The current Smart Cane prototype successfully demonstrates intelligent obstacle detection, fall detection, voice guidance, and emergency notification. However, as a prototype, it has several limitations that present opportunities for future improvement. Although the system can detect obstacles at ground and upper-body levels, it cannot detect downward hazards such as stairs, curbs, potholes, or open manholes because both ultrasonic sensors face forward. In addition, the emergency notification feature depends on the availability of a configured Wi-Fi network, limiting its effectiveness when the user is outside the network's coverage. The prototype also lacks battery monitoring, GPS location tracking, and customizable user settings such as alert distances and audio volume. These limitations reduce the overall reliability and practicality of the device for everyday use.

3.2 Next Steps

Following completion of this course, the project will continue to be developed into a more reliable and deployable assistive device. The team plans to integrate GPS and cellular communication for location-aware emergency alerts, implement downward-facing sensors for staircase and curb detection, add battery monitoring and power management features, and develop a companion mobile application for configuring device settings and monitoring system status. The project will also undergo extensive field testing with visually impaired users to evaluate usability, collect feedback, and improve the system before considering a production-ready version.

References

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4 Appendix A – Project Work plan

- **Week 1:** Project planning, problem identification, literature review, and selection of hardware components.
- **Week 2:** Procurement of components, hardware assembly, ESP32 setup, and integration of ultrasonic sensors.
- **Week 3:** Development of the embedded software, implementation of obstacle detection, fall detection, audio feedback, and Wi-Fi communication.
- **Week 4:** System integration, debugging, performance testing, and validation of all hardware and software modules.
- **Week 5:** Documentation, preparation of the technical report, final testing, and presentation of the completed prototype.

Appendix B – Contribution by Team Members

No.	Team Member	Contribution
1.	WASSWA KATEREGGA MAURICE	Team leader; coordinated the project, designed the system architecture, programmed the ESP32 firmware, integrated the ultrasonic sensors, MPU6050, DFPlayer Mini, and Wi-Fi notification system, managed the GitHub repository, and participated in testing and report writing.
2.	SSEKIZIYIVU RAYMOND	Assisted with hardware assembly, wiring of electronic components, sensor integration, prototype testing, debugging, and documentation of system functionality.
3.	MUYIMBA MICHEAL	Conducted literature review and requirements analysis, assisted in hardware setup, performed system testing and validation, and contributed to the technical report and presentation materials.
4.	SENNYANGE RODNEY	Assisted in prototype construction, power supply integration, testing of obstacle detection and fall detection features, troubleshooting hardware issues, and documenting project progress.
5.	NAMUTEBI SWABULAH	Participated in project planning, testing and evaluation of the prototype, documentation, report editing and formatting, and preparation of the final presentation.